

JEE MAINS QFT-20 2025



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I know the failure haunts you.... ✨



By: C I P H E R

with is
Saddam
doing
in QFT-20?

Saddam?
Hussein!

Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Body P having mass M moving with speed u has head-on collision elastically with another body Q having mass m initially at rest. If $m \ll M$, body Q will have a maximum speed equal to $2u$ after collision.

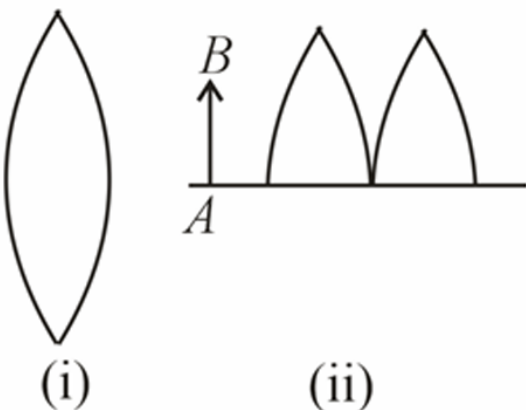
Reason R : During elastic collision, the momentum and kinetic energy are both conserved.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) A is correct but R is not correct. (B) A is not correct but R is correct.
- (C) Both A and R are correct and R is the correct explanation of A (D) Both A and R are correct but R is NOT the correct explanation of A

Question No. 2 (Only One Option Correct Type)

In the figure (i) a thin lens of focal length 10 cm is shown. The lens is cut into two equal parts, and the parts are arranged as shown in the figure (ii). An object AB of height 1 cm is placed at distance of 7.5 cm from the arrangement. The height of the final image will be: (in cm)



(A) 2.5 cm

(B) 2 cm

(C) 3.5 cm

(D) 3 cm

Question No. 3 (Only One Option Correct Type)

A mass 0.9 kg, attached to a horizontal spring, executes SHM with an amplitude A_1 . When this mass passes through its mean position, then a smaller mass of 124 g is placed over it and both masses move together with amplitude A_2 . If the ratio $\frac{A_1}{A_2}$ is $\frac{\alpha}{\alpha-1}$, then the value of α will be ____.

(A) 14

(B) 12

(C) 18

(D) 16

Question No. 4 (Only One Option Correct Type)

One mole of diatomic gas is being heated in a closed tank from 300 K up to 1000 K. During the process part of the molecules dissociate. At 1000 K the energy of the diatomic molecules are only half of that of the whole gas. By what factor has the gas pressure increased ($P_{\text{final}}/P_{\text{initial}}$)? (The oscillation of the molecules are not to be taken in account).

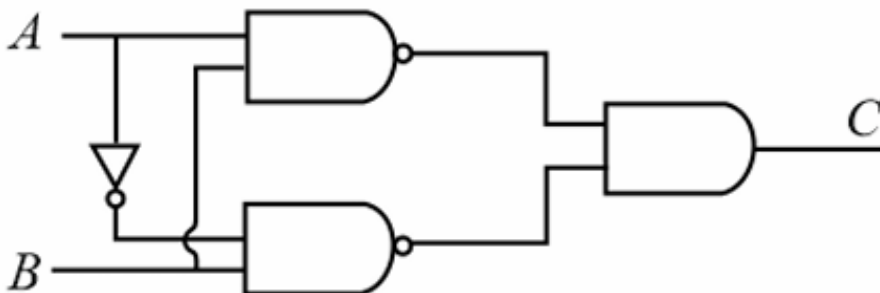
(A) $\frac{160}{33}$

(B) $\frac{16}{11}$

(C) $\frac{3}{2}$

(D) None

Question No. 5 (Only One Option Correct Type)



The truth table for the given logic circuit is:

	A	B	C
	0	0	1
(A)	0	1	0
	1	0	0
	1	1	1

	A	B	C
	0	0	1
(B)	0	1	0
	1	0	1
	1	1	0

	A	B	C
	0	0	1
(C)	0	1	1
	1	0	0
	1	1	1

	A	B	C
	0	0	0
(D)	0	1	1
	1	0	1
	1	1	0

Question No. 6 (**Only One Option Correct Type**)

The electron drift speed is small, and the charge of the electron is also small, but still, we obtain large current in a conductor. This is due to:

- (A) the conducting property of the conductor
- (B) the resistance of the conductor is small
- (C) the electron number density of the conductor is small
- (D) the electron number density of the conductor is enormous

Question No. 7 (**Only One Option Correct Type**)

Which of the following pairs of physical quantities does not have same dimensional formula ?

- (A) Work and torque
- (B) Angular momentum and Planck's constant
- (C) Tension and surface tension
- (D) Impulse and linear momentum

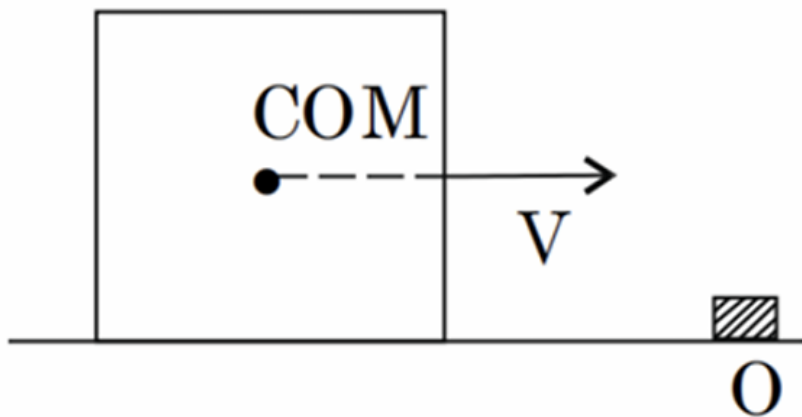
Question No. 8 (**Only One Option Correct Type**)

A body is moving unidirectionally under the influence of a source of constant power. The square of its displacement in time t is proportional to

- (A) t^3
- (B) t^2
- (C) t
- (D) $\sqrt{t}$

Question No. 9 (Only One Option Correct Type)

An ice cube of mass M and with sides of length a is sliding without friction with speed v_0 when it hits a ridge O at the edge of counter (see figure). This collision causes the cube to tilt as shown. The minimum value of v_0 needed for the cube to fall off the table is given by $\sqrt{kag}$. Find value of $10k$.



(A) 11

(B) 22

(C) 5.5

(D) 33

Question No. 10 (Only One Option Correct Type)

In a hydrogen sample, if the atoms are excited to states with principal quantum number 20, then the number of different wavelengths which may be observed in the spectrum is :

(A) 210

(B) 190

(C) 160

(D) 240

Question No. 11 (Only One Option Correct Type)

A convex lens of focal length 25 cm and made of glass with refractive index 1.5 is immersed in water. The absolute change in focal length of the glass is

[Use refractive index of water = $\frac{4}{3}$]

(A) 100 cm

(B) 37.5 cm

(C) 75 cm

(D) 12.5 cm

Question No. 12 (Only One Option Correct Type)

For photoelectric effect which of the following statements are true.

I) The kinetic energies of the photoelectrons do not depend on the frequency of light

II) Photoelectric effect will always occur for highly intense light

III) The maximum kinetic energy of photoelectron does not depend upon the intensity of the light

IV) The escaping electron's kinetic energy is larger for larger frequency

(A) I and II only

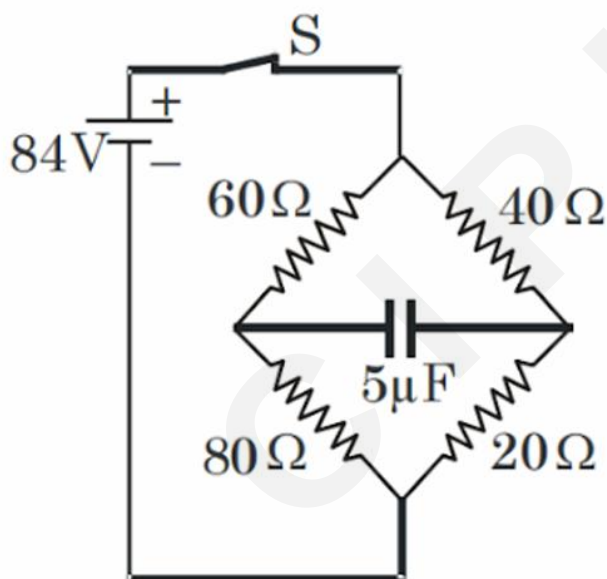
(B) II and III only

(C) III and IV only

(D) IV and I only

Question No. 13 (Only One Option Correct Type)

The circuit shown in the figure is in steady state for a long time. The connection to battery is suddenly broken (switch S is opened up). What is the charge (in μC) on the capacitor after 0.001 sec.?



(A) 100

(B) $100e^{-2}$

(C) $100e^{-4}$

(D) 0

Question No. 14 (Only One Option Correct Type)

A plane electromagnetic wave is propagating along the positive z direction. If the electric field component of this wave is in the direction $(\hat{i} + \hat{j})$, then which of the following is the direction of the magnetic field component?

(A) $(-\hat{i} + \hat{j})$

(B) $(\hat{i} - \hat{j})$

(C) $(-\hat{i} - \hat{j})$

(D) $(\hat{i} + \hat{k})$

Question No. 15 (Only One Option Correct Type)

It is known for a certain liquid that it takes 70 s to cool from 55°C to 50°C and 30 s to cool from 95°C to 90°C . Figure out the ambient temperature of the room.

(A) 16.5°C

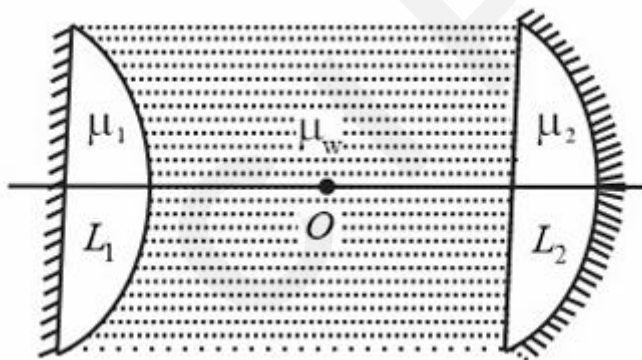
(B) 22.5°C

(C) 28.5°C

(D) 32.5°C

Question No. 16 (Only One Option Correct Type)

A cylindrical tube filled with water ($\mu_w = \frac{4}{3}$) is closed at its both ends by two thin, silvered plano-convex lenses, as shown in the figure. Refractive index of lenses L_1 and L_2 are 2.0 and 1.5, while their radii of curvatures are 5 cm and 9 cm, respectively. A point object is placed somewhere at a point O on the axis of the cylindrical tube. If it is found that all the images formed by multiple refractions and reflections coincide with the object, then the distance between both the lenses is



(A) 8 cm

(B) 18 cm

(C) 10 cm

(D) 14 cm

Question No. 17 (Only One Option Correct Type)

Consider a thermodynamic process where internal energy $U = AP^2 V$ ($A = \text{constant}$). If the process is performed adiabatically, then

(A) $AP^2(V + 1) = \text{constant}$

(B) $(AP + 1)^2 V = \text{constant}$

(C) $(AP + 1)V^2 = \text{constant}$

(D) $\frac{V}{(AP+1)^2} = \text{constant}$

Question No. 18 (Only One Option Correct Type)

Match the physical quantities with their correct SI units.

	List-I		List-II
(A)	Planck's constant	(I)	kg m s^{-2}
(B)	Torque	(II)	$\text{kg}^{-1} \text{ m}^3$
(C)	Weight	(III)	$\text{m}^2 \text{ kg s}^{-1}$
(D)	Specific volume	(IV)	$\text{kg m}^2 \text{ s}^{-2}$

The correct match is

(A) A = II, B = IV, C = I, D = III

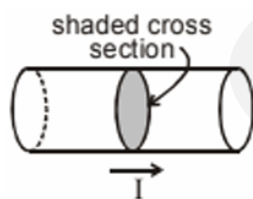
(B) A = III, B = IV, C = I, D = II

(C) A = IV, B = I, C = II, D = III

(D) A = II, B = III, C = I, D = IV

Question No. 19 (Only One Option Correct Type)

A current I flows through a cylindrical rod of uniform cross-section area A and resistivity ρ . The electric flux through the shaded cross-section of rod as shown in figure is:



(A) $\frac{I}{\rho}$

(B) ρI

(C) $\frac{\rho I}{A}$

(D) $\frac{\rho A}{I}$

Question No. 20 (Only One Option Correct Type)

Two point charges $-q$ and $+q$ are located at points $(0, 0, -a)$ and $(0, 0, a)$ respectively. The electric potential at a point $(0, 0, z)$, where $z > a$ is

(A) $\frac{qa}{4\pi\epsilon_0 z^2}$

(B) $\frac{q}{4\pi\epsilon_0 a}$

(C) $\frac{2qa}{4\pi\epsilon_0(z^2-a^2)}$

(D) $\frac{2qa}{4\pi\epsilon_0(z^2+a^2)}$

Physics Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

A closely wound solenoid of length 1 m has 5 layers of 500 turns each. If the magnitude of magnetic field inside the solenoid near its centre is 4.4 mT, the current carried is K, find 10K.

Question No. 2 (Numerical Type)

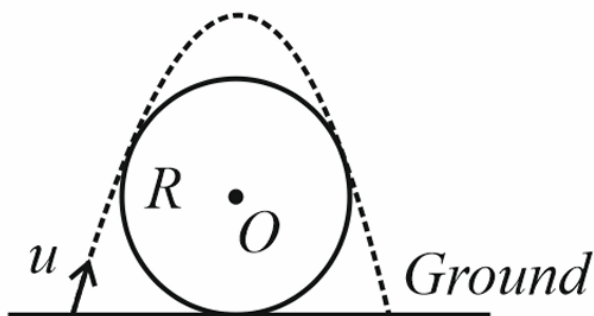
The plates of a parallel plate capacitor are charged upto 200 volts. A dielectric slab of thickness 4 mm is inserted between its plates. Then, to maintain the same potential difference between the plates of the capacitor, the distance between the plates is increased by 3.2 mm. The dielectric constant of the dielectric slab is

Question No. 3 (Numerical Type)

Binary stars rotate under mutual gravitational force at separation $2\left(\frac{G}{\omega^2}\right)^{\frac{1}{3}}$, where ω is the angular velocity of each of the star about centre of mass of the system. If difference between the mass of stars is 6 units. Find the ratio of mass of bigger star to smaller star.

Question No. 4 (Numerical Type)

One has to throw a particle from one side of a fixed sphere, in diametrical plane to another side such that it just grazes the sphere. Minimum possible speed for this is $\sqrt{2gR(\sqrt{\alpha} + \sqrt{\beta})}$. Find $\alpha + \beta$.



Question No. 5 (Numerical Type)

The relative density of the material of a body is the ratio of its weight in air and the loss of its weight in water. By using a spring balance, the weight of the body in the air is measured to be 5.00 ± 0.05 N. The weight of the body in water is measured to be 4.00 ± 0.05 N. Then the maximum possible percentage error in relative density is (mark answer in percentage)

Chemistry Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

Match the following

	Column-I		Column-II
(A)	$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$	(p)	Addition of inert gas at constant pressure shifts the equilibrium in the forward direction.
(B)	$2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$	(q)	Increase in pressure shifts the equilibrium in the forward direction
(C)	$\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$	(r)	$K_p = K_c$
(D)	$\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$	(s)	$K_p < K_c$

(A) A – QS, B – QS, C – P, D – R

(B) A – P, B – QS, C – QS, D – R

(C) A – P, B – QS, C – R, D – QS

(D) A – R, B – QS, C – P, D – QS

Question No. 2 (Only One Option Correct Type)

At some temperature, the rate constant for the decomposition of HI on a gold surface is 0.08Ms^{-1} . The reaction is $2\text{HI} \rightarrow \text{H}_2(\text{g}) + \text{I}_2(\text{g})$ The time taken for the concentration of HI to drop from 1.50M to 0.06M is _____ (sec).

(A) 6

(B) 9

(C) 12

(D) 15

Question No. 3 (Only One Option Correct Type)

Statement I : Liquid-liquid junction potential can be eliminated by putting a salt bridge of KCl in electrochemical cell.

Statement II : The function of salt bridge is to remove liquid-liquid junction potential because the salt used has same speed of cations and anions.

Statement I is true, Statement II is also

Statement I is true, Statement II is also

(A) true and Statement II is the correct explanation of Statement I.

(B) true and Statement II is not the correct explanation of Statement I.

(C) Statement I is true, Statement II is false.

(D) Statement I is false, Statement II is true.

Question No. 4 (Only One Option Correct Type)

Which of the following statements are correct?

i. $[\text{Mn}(\text{CN})_6]^{3-}$, $[\text{MnCl}_6]^{3-}$ both are paramagnetic

ii. Both $[\text{NiCl}_4]^{2-}$, $[\text{Ni}(\text{CO})_4]$ are tetrahedral and diamagnetic

iii. $[\text{Co}(\text{NH}_3)_6]^{3+}$ is inner orbital complex, but $[\text{Ni}(\text{NH}_3)_6]^{2+}$ is outer orbital complex

(A) i, ii only

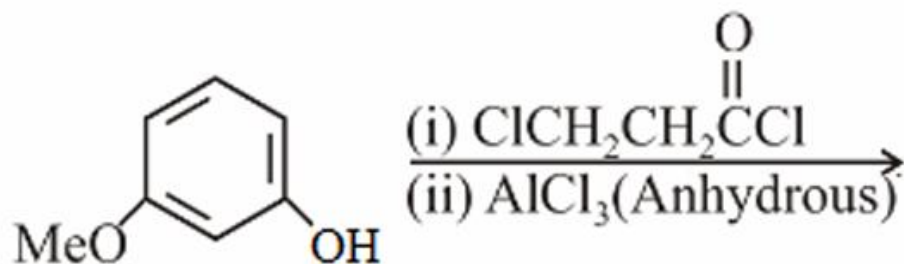
(B) i, iii only

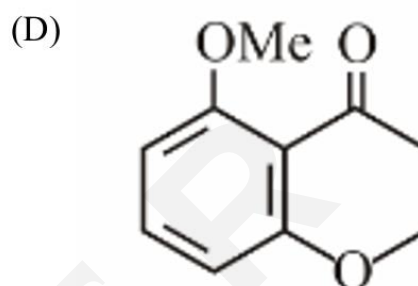
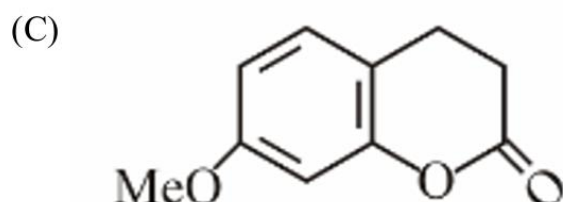
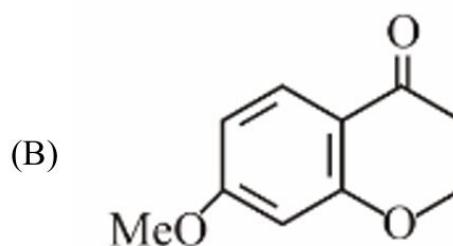
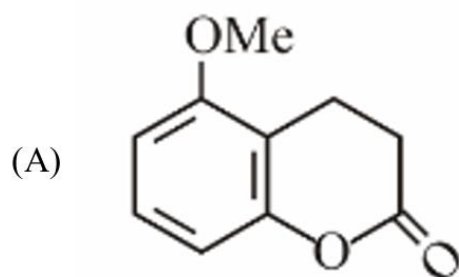
(C) ii, iii only

(D) i, ii, iii

Question No. 5 (Only One Option Correct Type)

The major product of the following reaction is:





Question No. 6 (Only One Option Correct Type)

Statement - 1: Only temperature can change the vapour pressure of a pure liquid.

Statement - 2: Equilibrium constant does not change unless temperature is changed.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

Question No. 7 (Only One Option Correct Type)

The correct statement among (a) to (d) regarding H_2 as a fuel are:

(a) It produces less pollutants than petrol

(b) A cylinder of compressed dihydrogen weight ~ 30 times more than a petrol tank producing the same amount of energy

(c) Dihydrogen is stored in tanks of metal alloys like $NaNi_5$

(d) On combustion, values of energy released per gram of liquid dihydrogen and LPG are 50 and 142 kJ, respectively

(A) (a)and(c)only

(B) (b), (c)and(d)only

(C) (a), (b) and (c) only

(D) (b) and (d) only

Question No. 8 (Only One Option Correct Type)

For emission line of atomic hydrogen from $n_i = 8$ to $n_f = n$, the plot of wave number $\left(\bar{\nu}\right)$ against $\left(\frac{1}{n^2}\right)$ will be: (The Rydberg constant, R_H is in wave number unit)

(A) Linear with slope $-R_H$

(B) Non linear

(C) Linear with slope R_H

(D) Linear with intercept $-R_H$

Question No. 9 (Only One Option Correct Type)

Assertion: Thiophene is present in commercial benzene as the impurity can be removed by shaking the mixture with cold concentrated H_2SO_4 .

Reason: Thiophene is a heterocyclic aromatic compound.

(A) Both assertion and reason are true, and the reason is the correct explanation of the assertion.

(B) Both assertion and reason are true, but the reason is not the correct explanation of the assertion.

(C) The assertion is true, but the reason is false.

(D) Both the assertion and the reason are false.

Question No. 10 (Only One Option Correct Type)

Which among the following statements are true for glycine?

1. It exists in crystalline form

2. It is optically active

3. It is soluble in water

4. It can form Zwitter ions

(A) 1, 2 and 3

(B) 1, 2 and 4

(C) 1, 3 and 4

(D) 2, 3 and 4

Question No. 11 (Only One Option Correct Type)

In the crystal field of the complex $[\text{Fe}(\text{Cl})(\text{CN})_4(\text{O}_2)]^{4-}$ the electronic configuration of metal is found to be t_{2g}^6, e_g^0 then which of the following is true about this complex ion :

- (A) All ligands are weak field ligands
- (B) $O - O$ bond length will be less than found in (O_2) molecule
- (C) Its IUPAC name will be chlorotetracyanosuperoxidoferrate (II) ion
- (D) It is a diamagnetic complex

Question No. 12 (Only One Option Correct Type)

Which of the following statements are correct for classification of elements?

- I. The properties of elements are periodic functions of their atomic numbers.
- II. Non-metallic elements are less in number than the metallic elements.
- III. The first ionisation energies of elements along a period do not vary in a regular manner.
- IV. The ground state electronic configuration of $\text{Pd}(Z = 46)$ is $[\text{Kr}]4d^85s^2$.

- (A) I, II, III, IV
- (B) I, II, III only
- (C) II, III, IV only
- (D) I, II, IV only

Question No. 13 (Only One Option Correct Type)

Which one of the following has maximum number of hybrid orbitals?

- (A) C_6H_6
- (B) $(\text{CH}_3)_4\text{C}$
- (C) $(\text{CH}_3)_2\text{C} = \text{O}$
- (D) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CN}$

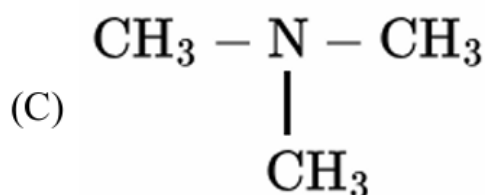
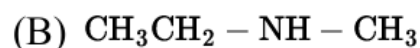
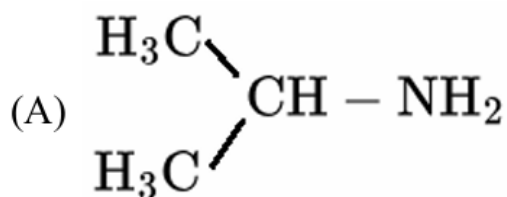
Question No. 14 (Only One Option Correct Type)

How many total number of elements (upto atomic number = 10) are here, which have positive electron gain enthalpy (EGE_1)?

- (A) 3
- (B) 4
- (C) 5
- (D) 6

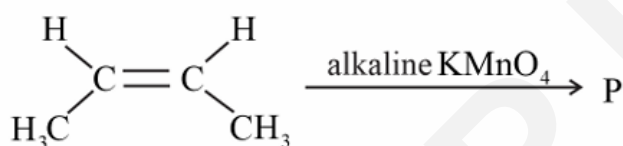
Question No. 15 (Only One Option Correct Type)

An organic compound (C_3H_9N)(A), when treated with nitrous acid, gave an alcohol and N_2 gas was evolved. (A) on warming with $CHCl_3$ and caustic potash gave (C) which on reduction gave isopropylmethanamine. Predict the structure of (A).



Question No. 16 (Only One Option Correct Type)

Which of the following statements is true for the reaction given below?



(A) P is a the meso compound 2, 3-butanediol formed by syn-addition

(B) P is a the meso compound 2, 3-butanediol formed by anti-addition

(C) P is a racemic mixture of (d) and (l) 2, 3-butanediol formed by anti-addition

(D) P is a racemic mixture of (d) and (l) 2, 3-butanediol formed by syn-addition

Question No. 17 (Only One Option Correct Type)

Which one is the wrong statement?

(A) de-Broglie's wavelength $\lambda = \frac{h}{mv}$ where m is the mass of the particle, and v is the velocity of the particle.

(B) The uncertainty principle is

$$\Delta x \times \Delta p \geq \frac{h}{4\pi}$$

(C) Half-filled and fully-filled orbitals have greater stability due to greater exchange energy, greater symmetry, and a more balanced arrangement.

(D) The energy of the **2s** orbital is less than the energy of the **2p** orbital in the case of hydrogen-like atoms.

Question No. 18 (Only One Option Correct Type)

Observe the following reaction



Identify the correct statements about this reaction

- I. $\text{Cl}_2(\text{g})$ is oxidized to $\text{ClO}^-(\text{aq})$
- II. $\text{Cl}_2(\text{g})$ is oxidized to $\text{Cl}^-(\text{aq})$
- III. $\text{Cl}_2(\text{g})$ is reduced to $\text{ClO}^-(\text{aq})$
- IV. $\text{Cl}_2(\text{g})$ is reduced to $\text{Cl}^-(\text{aq})$

The correct answer is

(A) I, IV only

(B) I, III only

(C) I, II, IV only

(D) II, III only

Question No. 19 (Only One Option Correct Type)

Assertion: Refining of petroleum involves the fractional distillation.

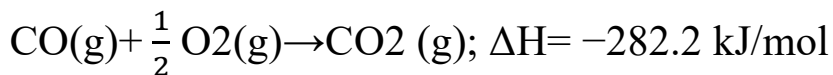
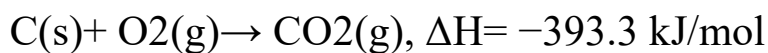
Reason: The fractional distillation involves repeated distillation. Repeated distillation means repeated heating.

- (A) Both assertion and reason are true, and the reason is the correct explanation of the assertion.
- (B) Both assertion and reason are true, but the reason is not the correct explanation of the assertion.
- (C) The assertion is true, but the reason is false.
- (D) Both the assertion and the reason are false.

Question No. 20 (Only One Option Correct Type)

Mark out the enthalpy of formation of carbon monoxide (CO).

Given,



(A) $-111.1 \text{ KJ mol}^{-1}$

(B) $676.1 \text{ KJ mol}^{-1}$

(C) $-282.8 \text{ KJ mol}^{-1}$

(D) $300.0 \text{ KJ mol}^{-1}$

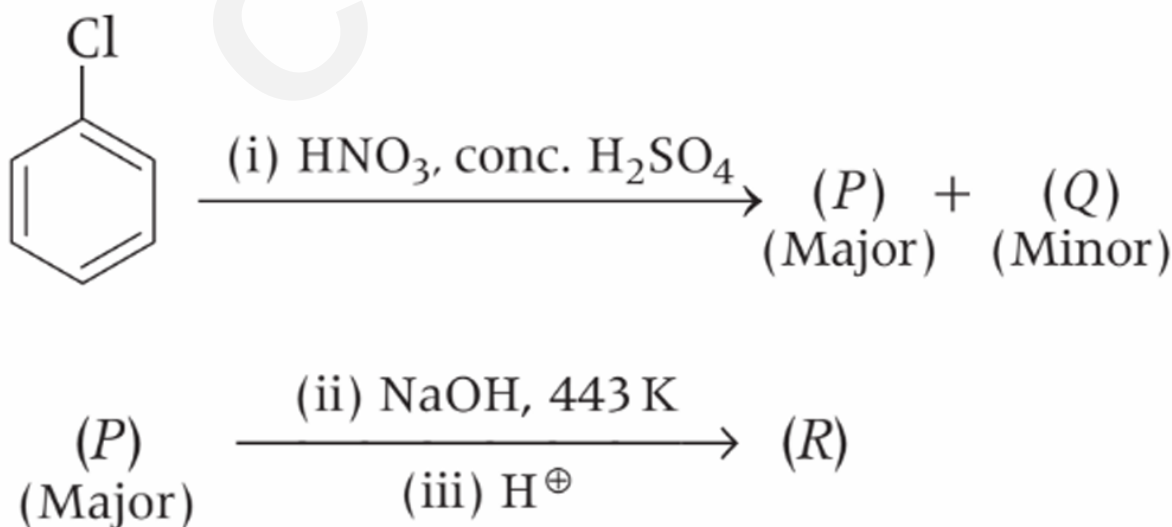
Chemistry Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

During the Kjeldahl method, 2.8 g of an organic compound was digested and the evolved ammonia was absorbed by 60 mL of $\frac{M}{20} \text{H}_2\text{SO}_4$. What volume of $\frac{M}{20} \text{NaOH}$ solution was required for complete neutralisation of the unreacted acid if the percentage of nitrogen in the compound was 1% ? (mark answer in mL)

Question No. 2 (Numerical Type)

The product (*R*) in the following synthetic scheme is



Find molar mass of product *R*.

Question No. 3 (Numerical Type)

Consider the following ligands

NH_2^- , acac, SH^- , Gly, O_2^- , Phen, DMG, NO_2^- , CO_3^{2-} , I^- , CH_3COO^- , en, SO_4^{2-} .

Then calculate value of " $P + Q - R - S$ "

Here, P: Total number of ligands which act as bridging as well as monodentate only.

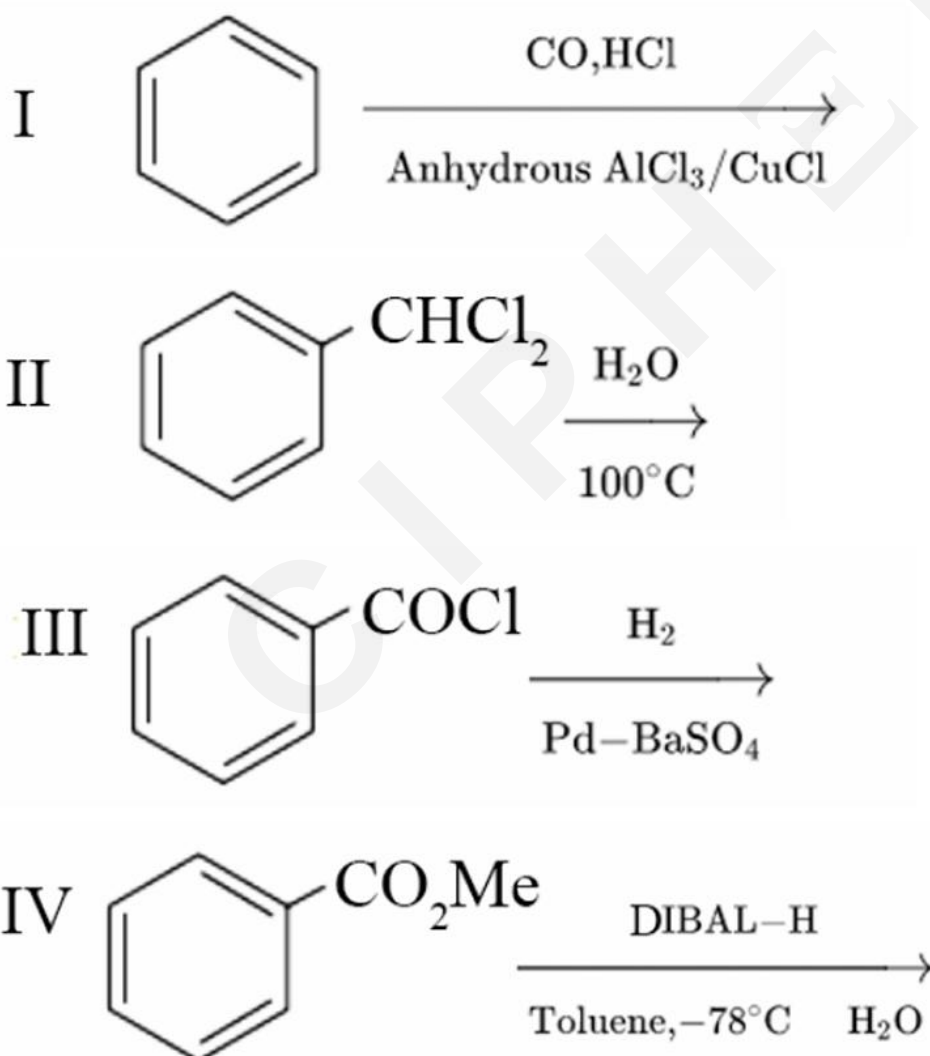
Q: Total number of flexidentate ligands

R: Total number of bidentate ligands only

S: Total number of unsymmetrical bidentate ligands

Question No. 4 (Numerical Type)

Among the following, the number of reaction(s) that produce(s) benzaldehyde is:



Question No. 5 (Numerical Type)

Concentration of NH_4Cl and NH_4OH in a buffer solution is in the ratio of 1 : 1, K_b for NH_4OH is 10^{-10} . The pH of the buffer is

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

Let $\vec{a} = \hat{i} + x\hat{j}$ & $\vec{b} = x\hat{j} + \hat{k}$, $x \in (0, 1)$. If $G(\vec{a} \cdot \vec{b}) = \int_0^{(\vec{a} \cdot \vec{b})} e^{\left[(\vec{a} \cdot \vec{b}) - \frac{(\vec{a} \cdot \vec{b})^2}{2} + \frac{(\vec{a} \cdot \vec{b})^3}{3} - \frac{(\vec{a} \cdot \vec{b})^4}{4} + \dots \dots \dots \infty\right]} d(\vec{a} \cdot \vec{b})$, then value of $\frac{d^2 G(\vec{a} \cdot \vec{b})}{d(\vec{a} \cdot \vec{b})^2}$ is:

(A) 0

(B) 1

(C) x

(D) $1 + x^2$

Question No. 2 (Only One Option Correct Type)

Given that $\vec{a}, \vec{b}, \vec{p}, \vec{q}$ are four vectors such that $\vec{a} + \vec{b} = \mu \vec{p}$, $\vec{b} \cdot \vec{q} = 0$ and $(\vec{b})^2 = 1$, where μ is a scalar then $|(\vec{a} \cdot \vec{q})\vec{p} - (\vec{p} \cdot \vec{q})\vec{a}|$ is equal to.....

(A) $2|\vec{p} \cdot \vec{q}|$

(B) $(1/2)|\vec{p} \cdot \vec{q}|$

(C) $|\vec{p} \times \vec{q}|$

(D) $|\vec{p} \cdot \vec{q}|$

Question No. 3 (Only One Option Correct Type)

Solution of differential equation $x^2 \frac{dy}{dx} + y^2 e^{\frac{x(y-x)}{y}} = 2y(x-y)$ be given by

Where C is constant of integration-

(A) $x(x+y) = y \log(Ce^x - 1)$

(B) $x(x-y) = x \log(Ce^x - 1)$

(C) $x(x+y) = x \log(Ce^x + 1)$

(D) $x(x-y) = y \log(Ce^x - 1)$

Question No. 4 (Only One Option Correct Type)

A square $ABCD$ of side length $\sqrt{50}$ is folded along diagonal AC so that planes ACB and ACD are perpendicular to one another where B' is the new position of B . If the shortest distance between AB' and CD is $\frac{10}{\sqrt{n}}$, then n is

(A) 2

(B) 5

(C) 3

(D) 8

Question No. 5 (Only One Option Correct Type)

Let the matrices $A = \begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix}$ and P be any orthogonal matrix such that $Q = PAP'$ and $R = [r_{ij}]_{2 \times 2} = P'Q^8P$ then.

(A) $r_{11} = 81$

(B) $r_{11} = 81\sqrt{3}$

(C) $r_{11} = 4\sqrt{3}$

(D) $r_{11} = -\sqrt{3}$

Question No. 6 (Only One Option Correct Type)

A rectangle PQRS has sides $PQ = 11$ and $QR = 5$. A triangle ABC has P as orthocentre, Q as circumcentre, R as mid point of BC and S as the foot of altitude from A. Then length of BC is k, where $k/4$ is equal to....

(A) 6

(B) 5

(C) 3

(D) 7

Question No. 7 (Only One Option Correct Type)

Six letter words are formed using all the letters of the word NUMBER. If these words are arranged in the dictionary order, the rank of the word NUMBER is?

(A) 467

(B) 468

(C) 469

(D) 470

Question No. 8 (Only One Option Correct Type)

The equations $x^2 + ax + 12 = 0$, $x^2 + bx + 15 = 0$ and $x^2 + (a + b)x + 36 = 0$ have a common positive root then the value of $a - b$ is---

(A) 1

(B) 2

(C) 3

(D) 4

Question No. 9 (Only One Option Correct Type)

Question No. 10 (Only One Option Correct Type)

If $f(x) = \begin{cases} (p^2 - 1)(\{x\} + 2[x]) - 2, & -2 < x \leq -1 \\ q\left(\frac{e^x + e^{-x}}{2}\right) + |p|(x - 1), & -1 < x < 2 \end{cases}$, $p, q \in R$ is continuous in $(-2, 2)$ then

$f(f(f(\frac{-1}{2})))$ is

[Note: $[k]$ denotes greatest integer function less than or equal to k and $\{k\}$ denotes fractional part function of k .]

(A) -2

(B) -1

(C) 0

(D) not defined

Question No. 11 (Only One Option Correct Type)

The area of the region $R = \{(x, y) : |x| \leq |y| \text{ and } x^2 + y^2 \leq 1\}$ is....

(A) $\frac{3\pi}{8}$ sq units

(B) $\frac{5\pi}{8}$ sq units

(C) $\frac{\pi}{2}$ sq units

(D) $\frac{\pi}{8}$ sq unit

Question No. 12 (Only One Option Correct Type)

The remainder on dividing $1234^{567} + 89^{1011}$ by 12 is

(A) 1

(B) 5

(C) 9

(D) none of these

Question No. 13 (Only One Option Correct Type)

A chord is drawn passing through $P(2, 2)$ on the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ such that it intersects the ellipse at points A and B . Then the maximum value of $PA \cdot PB$ is equal to

(A) $\frac{61}{4}$

(B) $\frac{59}{4}$

(C) $\frac{71}{4}$

(D) $\frac{63}{4}$

Question No. 14 (Only One Option Correct Type)

Consider two sets

$$A = \left\{ \frac{p}{q}; p, q \text{ are coprime odd integers} \right\} \text{ and}$$

$$B = \left\{ \frac{p+1}{q} : p, q \text{ are odd integers and } p \neq -1 \right\}.$$

Functions $f : A \rightarrow \mathbb{R}$ & $g : B \rightarrow \mathbb{R}$ are both defined as $f(x) = x^x$ & $g(x) = x^{|x|}$ on respective domains.

Then mark incorrect option

(A) $f\left(\frac{1}{3}\right) \cdot g\left(\frac{-2}{3}\right) = \frac{\sqrt[3]{4}}{3}$

(B) $g(x)$ is an even function.

(C) $f(1) + g(2) = 6$

(D) $\left[g\left(\frac{-6}{5}\right)\right]^{5/2} - \left[g\left(\frac{4}{5}\right)\right]^{15/4} - \left[f\left(\frac{3}{5}\right)\right]^5 = 1$

Question No. 15 (Only One Option Correct Type)

A circle with centre at $(2, 4)$ is such that the line $x + y + 2 = 0$ cuts a chord of length 6 units. The radius of the circle is

(A) $\sqrt{41}$ units

(B) $\sqrt{11}$ units

(C) $\sqrt{21}$ units

(D) $\sqrt{31}$ units

Question No. 16 (Only One Option Correct Type)

In a quadrilateral $ABCD$, $\overrightarrow{AC}$ is the bisector of $\overrightarrow{AB}$ and $\overrightarrow{AD}$, angle between $\overrightarrow{AB}$ and $\overrightarrow{AD}$ is $\frac{2\pi}{3}$, $15|\overrightarrow{AC}| = 3|\overrightarrow{AB}| = 5|\overrightarrow{AD}|$. Then the angle between $\overrightarrow{BA}$ and $\overrightarrow{CD}$ is

(A) $\cos^{-1} \frac{\sqrt{14}}{7\sqrt{2}}$

(B) $\cos^{-1} \frac{\sqrt{21}}{7\sqrt{3}}$

(C) $\cos^{-1} \frac{2}{\sqrt{7}}$

(D) $\cos^{-1} \frac{2\sqrt{7}}{14}$

Question No. 17 (Only One Option Correct Type)

A locker can be opened by dialing a fixed three digit code (between 000 and 999). A stranger who does not know the code tries to open the locker by dialing three digits at random. The probability that the stranger succeeds at the k^{th} trial is

(A) $\frac{k}{999}$

(B) $\frac{k}{1000}$

(C) $\frac{k-1}{1000}$

(D) None of these

Question No. 18 (Only One Option Correct Type)

The number of values of α in $[-10\pi, 10\pi]$ for which the equations $(\sin \alpha)x - (\cos \alpha)y + 3z = 0$, $(\cos \alpha)x + (\sin \alpha)y - 2z = 0$ and $2x + 3y + (\cos \alpha)z = 0$ have a nontrivial solution is

(A) 10

(B) 20

(C) 40

(D) 15

Question No. 19 (Only One Option Correct Type)

There are 100 sets named $A_1, A_2, \dots, A_{100}$ such that $n(A_i) = i + 2$ represents the number of elements in A_i and $A_1 \subset A_2 \subset A_3 \dots \subset A_{100}$. If $\bigcap_{i=3}^{100} A_i = A$ then $n(A) =$

(A) 3

(B) 4

(C) 5

(D) 6

Question No. 20 (Only One Option Correct Type)

The value of $\sin \frac{9\pi}{14} \sin \frac{11\pi}{14} \sin \frac{13\pi}{14}$ is equal to

(A) $\frac{1}{64}$

(B) $-\frac{1}{64}$

(C) $\frac{1}{8}$

(D) $-\frac{1}{8}$

Mathematics Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

If $I_n = \int_0^{n\pi} \max(|\sin x|, |\sin^{-1}(\sin x)|) dx$, then $I_2 + I_4$ has the value $\frac{\lambda\pi^2}{2}$; where λ is

Question No. 2 (Numerical Type)

Let $a_1, a_2, a_3, \dots, a_{49}$ be in A.P. such that

$$\sum_{k=0}^{12} a_{4k+1} = 416$$

and

$$a_9 + a_{43} = 66$$

If

$$a_1^2 + a_2^2 + \dots + a_{17}^2 = 140m,$$

then m is equal to

$$\left(\text{given that } 1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} \right)$$

Question No. 3 (Numerical Type)

$$\lim_{h \rightarrow \infty} h^{-h^2} \left[(h+1) \left(h + \frac{1}{2} \right) \left(h + \frac{1}{2^2} \right) \dots \left(h + \frac{1}{2^{h-1}} \right) \right]^h = e^S$$

then $S = \dots$

Question No. 4 (Numerical Type)

If the line $y - 2 = 0$ is the directrix of the parabola $x^2 - ky + 32 = 0, k \neq 0$ and the parabola intersects the circle $x^2 + y^2 = 8$ at two real distinct points, then the absolute value of k is

Question No. 5 (Numerical Type)

If ω is the imaginary cube root of unity, then find the number of pairs of integers (a, b) such that $|a\omega + b| = 1$

(ANSWER KEY)

Physics

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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Chemistry

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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Mathematics

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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